

Project Development Phase – Part 1

1.Data Collection:

The dataset used for this project was provided through the **SkillWallet Internship Platform**. The dataset is named **Heart_new2** and contains healthcare information used for analyzing heart disease risk factors.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	HeartDisease	BMI	Smoking	AlcoholDrin	Stroke	PhysicalHea	MentalHeal	DiffWalking	Sex	AgeCategor	Race	Diabetic	PhysicalActi	GenHealth	SleepTime	Asthma	KidneyDise:S
2	No	16.6	Yes	No	No	3	30	No	Female	55-59	White	Yes	Yes	Very good	5	Yes	No
3	No	20.34	No	No	Yes	0	0	No	Female	80 or older	White	No	Yes	Very good	7	No	No
4	No	26.58	Yes	No	No	20	30	No	Male	65-69	White	Yes	Yes	Fair	8	Yes	No
5	No	24.21	No	No	No	0	0	No	Female	75-79	White	No	No	Good	6	No	No
6	No	23.71	No	No	No	28	0	Yes	Female	40-44	White	No	Yes	Very good	8	No	No
7	Yes	28.87	Yes	No	No	6	0	Yes	Female	75-79	Black	No	No	Fair	12	No	No
8	No	21.63	No	No	No	15	0	No	Female	70-74	White	No	Yes	Fair	4	Yes	No
9	No	31.64	Yes	No	No	5	0	Yes	Female	80 or older	White	Yes	No	Good	9	Yes	No
10	No	26.45	No	No	No	0	0	No	Female	80 or older	White	No, borderl	No	Fair	5	No	Yes
11	No	40.69	No	No	No	0	0	Yes	Male	65-69	White	No	Yes	Good	10	No	No
12	Yes	34.3	Yes	No	No	30	0	Yes	Male	60-64	White	Yes	No	Poor	15	Yes	No
13	No	28.71	Yes	No	No	0	0	No	Female	55-59	White	No	Yes	Very good	5	No	No
14	No	28.37	Yes	No	No	0	0	Yes	Male	75-79	White	Yes	Yes	Very good	8	No	No
15	No	28.15	No	No	No	7	0	Yes	Female	80 or older	White	No	No	Good	7	No	No
16	No	29.29	Yes	No	No	0	30	Yes	Female	60-64	White	No	No	Good	5	No	No
17	No	29.18	No	No	No	1	0	No	Female	50-54	White	No	Yes	Very good	6	No	No
18	No	26.26	No	No	No	5	2	No	Female	70-74	White	No	No	Very good	10	No	No
19	No	22.59	Yes	No	No	0	30	Yes	Male	70-74	White	No, borderl	Yes	Good	8	No	No
20	No	29.86	Yes	No	No	0	0	Yes	Female	75-79	Black	Yes	No	Fair	5	No	Yes
21	No	18.13	No	No	No	0	0	No	Male	80 or older	White	No	Yes	Excellent	8	No	No

2. Loading Dataset into Tableau:

The **Heart_new2** dataset was imported into Tableau Desktop for visualization and analysis. Tableau automatically detected the fields and assigned appropriate data types.

Tableau Public - Heart_Disease_Analysis

FileDataWindowHelp

Connections

Heart_new2

Text file

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

Advertising.csv

Attendance_Dataset.csv

claimants (1).csv

CO2 Emissions_Canada.csv

data.csv

diabetes.csv

gene_expression.csv

healthcare-data...stroke-data.csv

Heart_new2.csv

mouse_viral_study.csv

penguins_size.csv

random_sample.csv

New Union

New Table Extension

Heart_new2

Heart_new2.csv

Heart_new2.csv

22 fields 4500 rows

100 rows

NameHeart_new2.csv

Description

No description available.

Fields

Type	Field Name	Phys...	Rem...
Abc	Heart Disease	Heart...	Heart...
#	BMI	Heart...	BMI
Abc	Smoking	Heart...	Smoki...

Heart Disease

BMI

Smoking

Alcohol Drinking

Stroke

Physical

No	16.6000	Yes	No	No	
No	20.3400	No	No	Yes	
No	26.5800	Yes	No	No	
No	24.2100	No	No	No	
No	23.7100	No	No	No	
Yes	28.8700	Yes	No	No	
No	21.6300	No	No	No	
No	31.6400	Yes	No	No	

Data Source

Heart Disease

General Health vs Heart Disease

Physical Activity vs Heart Disease

Age and BMI vs Diabetic

People Got Stroke Suffering fro...

Story 1

Dashboard 3

Dashboard 4

3. Data Preparation:

The following preprocessing steps were performed before creating visualizations.

Data Preparation Steps

- Imported the Heart_new2 dataset into Tableau.
- Verified all field names.
- Checked data types.
- Identified Dimensions and Measures.
- Created calculated fields.
- Verified the dataset before visualization.

Calculated Fields Created

- Heart Disease Count
- Heart Disease Rate
- Current Smokers

ADC	SMOKING
Abc	Stroke
Abc	<i>Measure Names</i>
#	BMI
#	Current Smokers
#	Heart Disease count
#	Heart Disease Rate
#	Mental Health
#	Physical Health
#	Sleep Time
#	<i>Heart_new2.csv (Count)</i>
#	<i>Measure Values</i>

Heart Disease Rate

×

```
SUM([Heart Disease count])/COUNT([Heart Disease])
```

The calculation is valid.

2 Dependencies ▾

Apply

OK

Current Smokers

×

```
IF [Smoking]=="Yes" THEN 1 END
```

The calculation is valid.

2 Dependencies ▾

Apply

OK

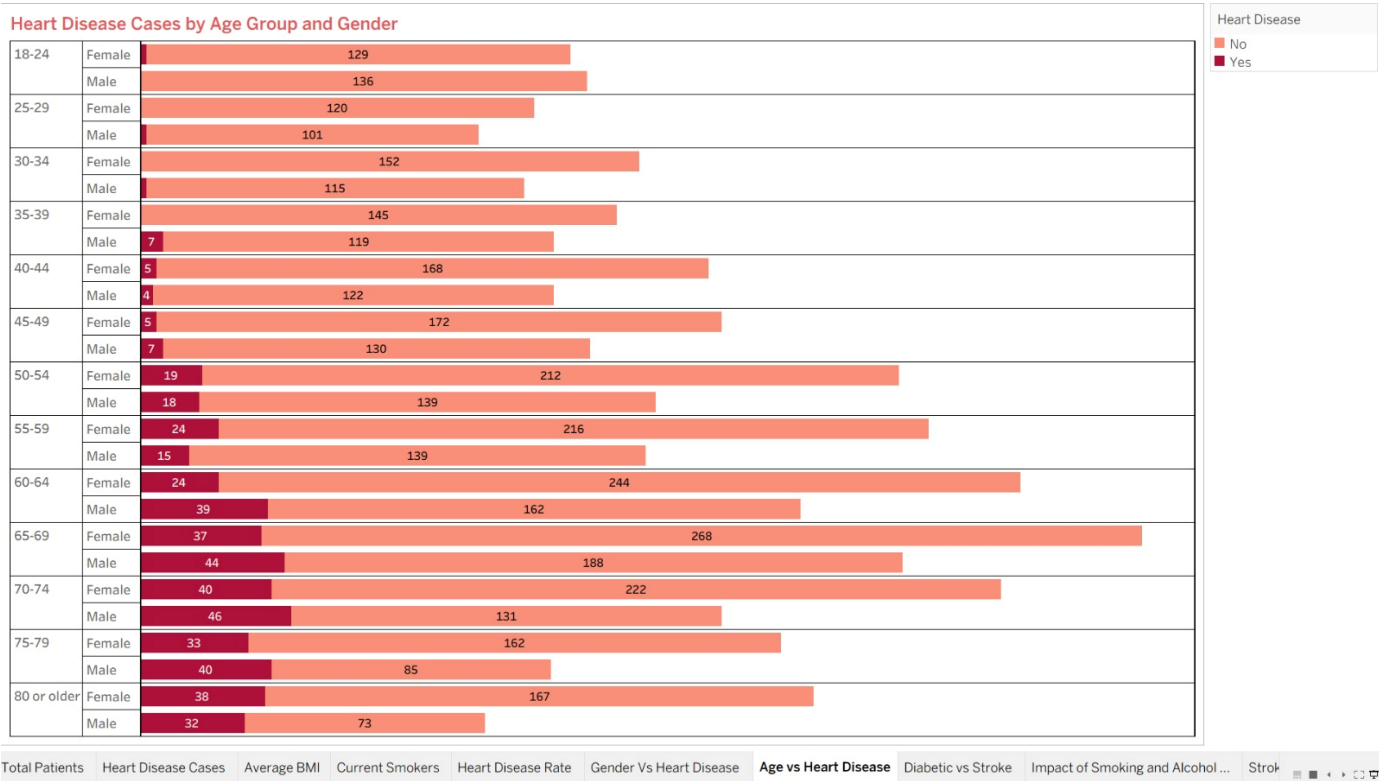
4. Business Questions with Visualizations

Business Question 1

How does heart disease vary across different age groups?

Visualization

Age vs Heart Disease



Insight

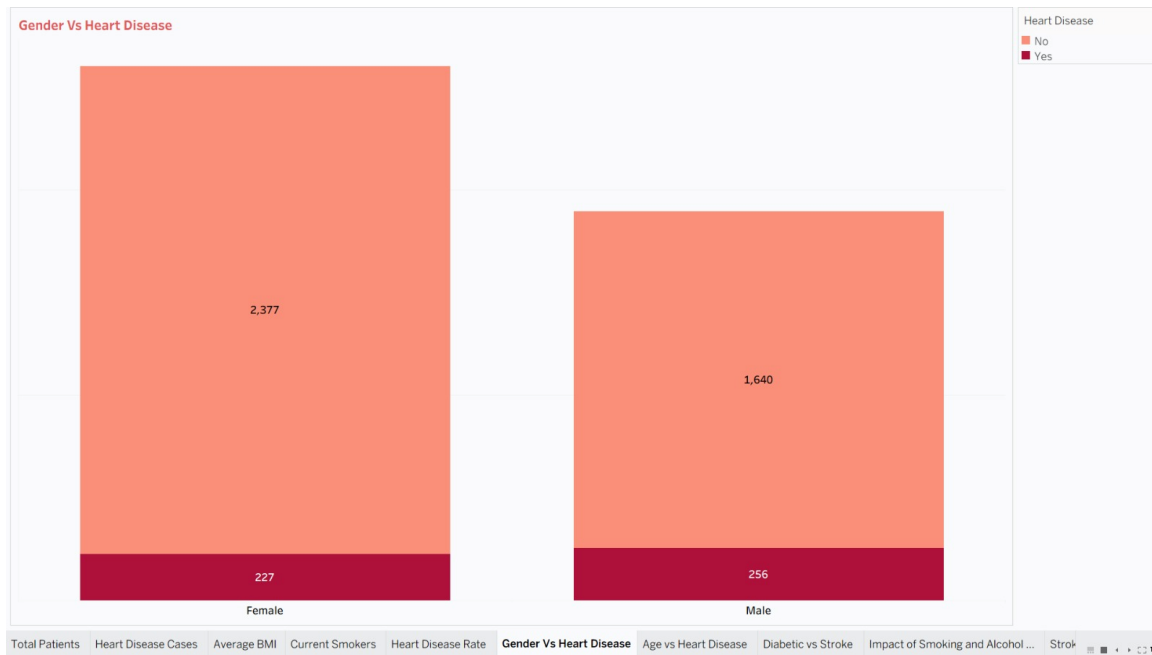
The visualization shows that the number of heart disease cases generally increases with age. Older age groups have a higher prevalence of heart disease compared to younger groups. This indicates that age is an important factor influencing the risk of developing heart disease.

Business Question 2

Which gender is more affected by heart disease?

Visualization

Gender vs Heart Disease



Insight

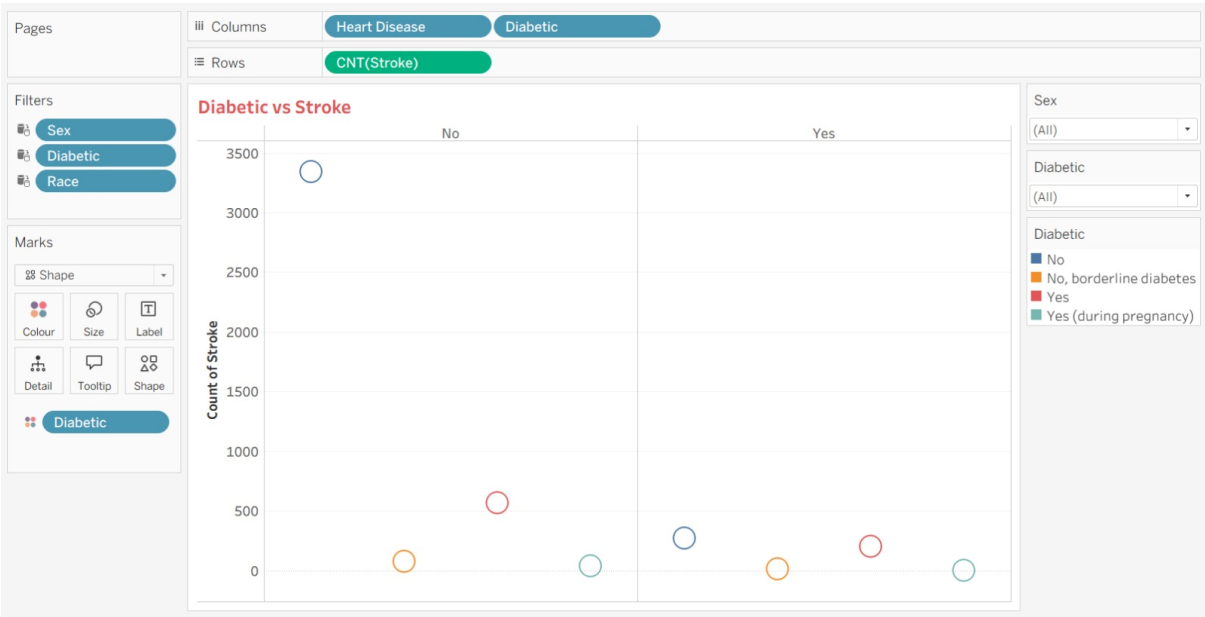
The chart compares heart disease cases between male and female patients. It helps identify differences in disease occurrence across genders and supports demographic analysis. Such insights can help in understanding population-specific health trends

Business Question 3

What is the relationship between diabetes and stroke?

Visualization

Diabetic vs Stroke



Insight

This visualization highlights the association between diabetes and stroke among patients. It helps identify how frequently stroke occurs in diabetic individuals compared to non-diabetic individuals. Understanding this relationship supports better health risk assessment.

Business Question 4

How do smoking and alcohol consumption affect heart disease?

Visualization

Impact of Smoking and Alcohol on Heart Disease



Insight

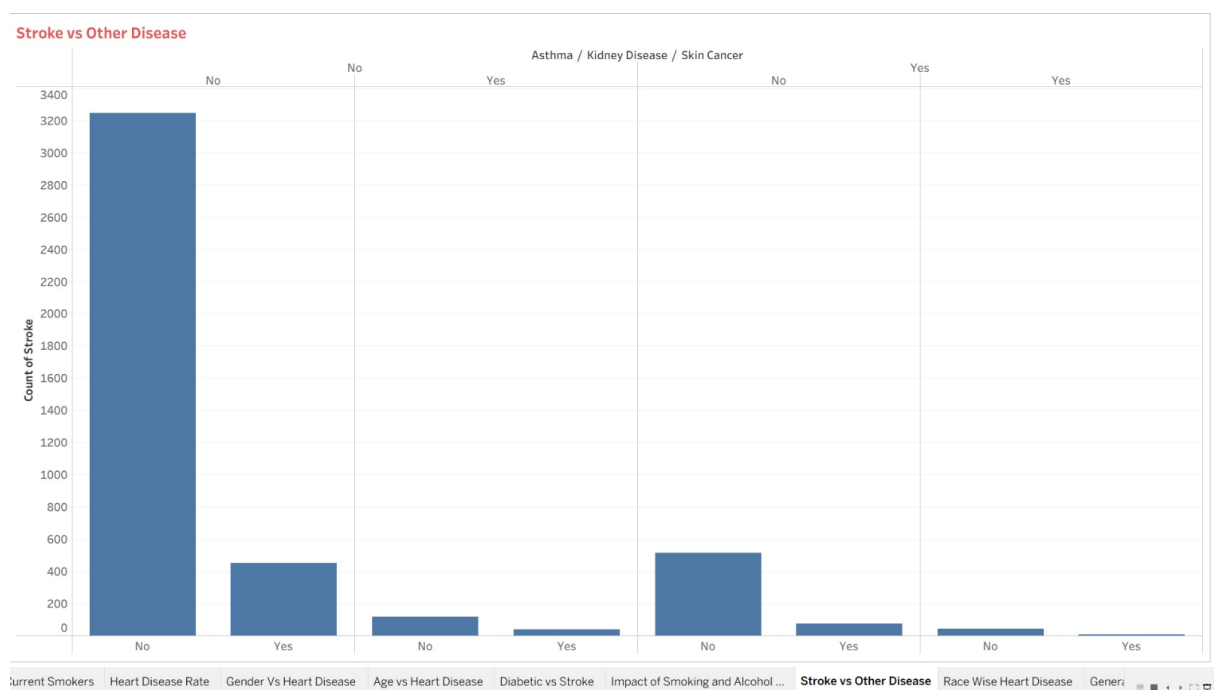
The chart illustrates the effect of smoking and alcohol consumption on heart disease occurrence. It enables comparison between different lifestyle groups and helps identify behavioral factors associated with heart disease. These insights can support awareness and preventive healthcare.

Business Question 5

How do other health conditions influence stroke?

Visualization

Other Health Diseases vs Stroke



Insight

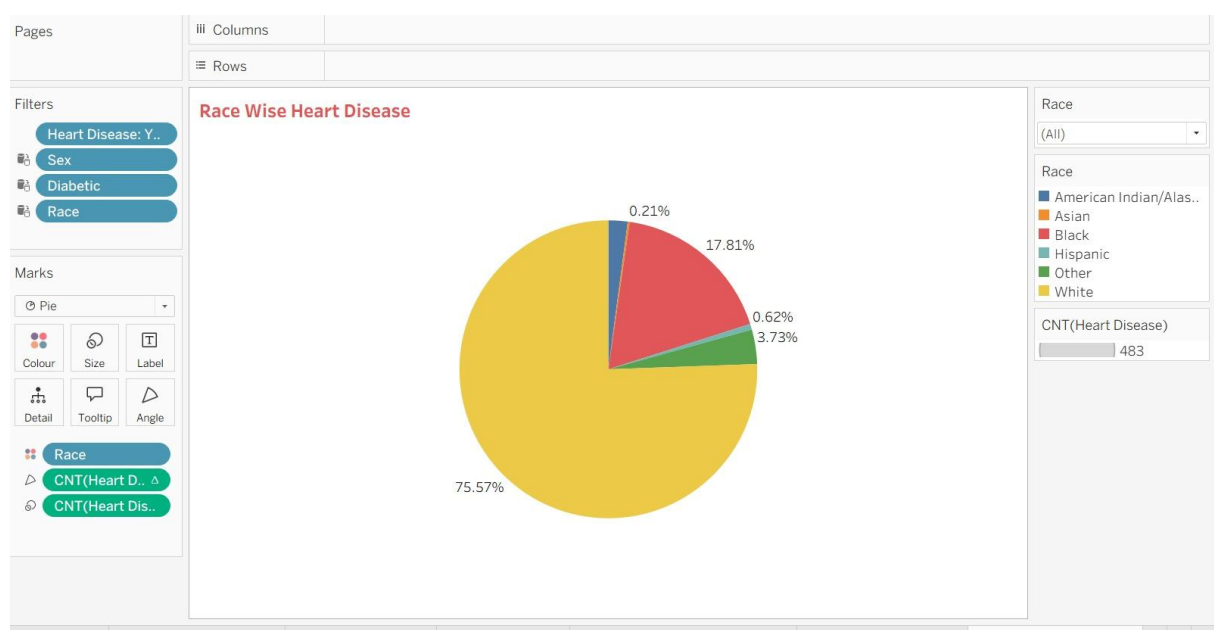
This visualization compares stroke occurrence among individuals with different health conditions such as asthma, kidney disease, and skin cancer. It helps identify health conditions that frequently coexist with stroke. Such analysis supports a broader understanding of patient health profiles.

Business Question 6

How is heart disease distributed among different races?

Visualization

Race-wise Heart Disease



Insight

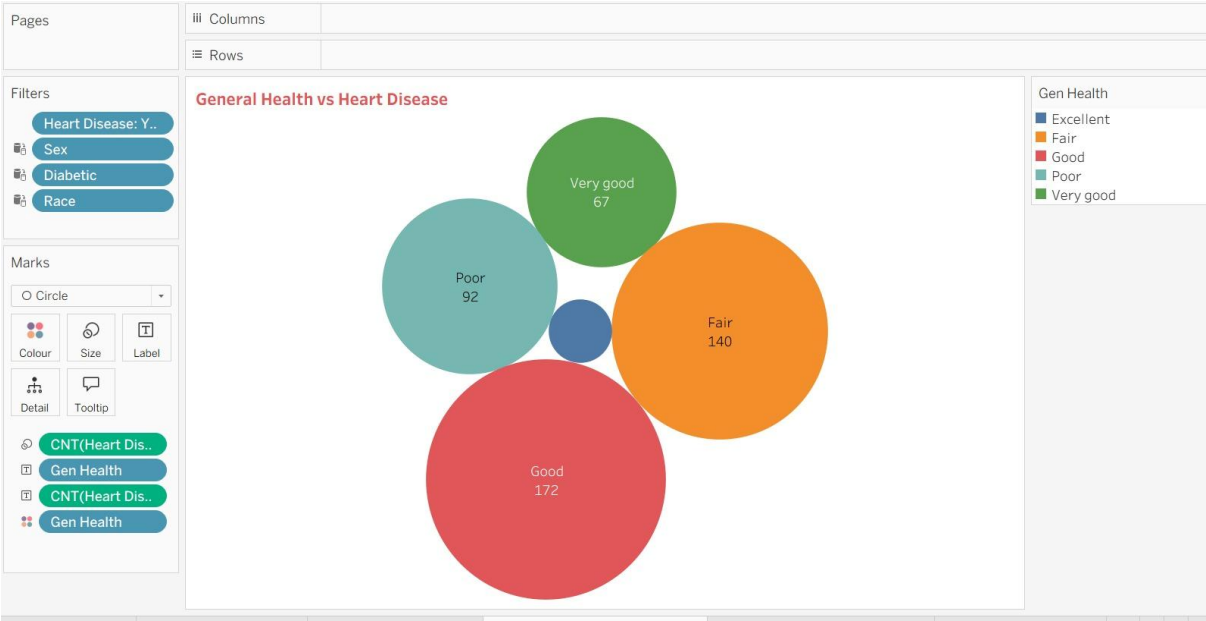
The visualization presents the distribution of heart disease across different racial groups in the dataset. It allows users to compare the number of heart disease cases among various populations. This comparison helps identify demographic patterns within the available data.

Business Question 7

Does general health status influence heart disease?

Visualization

General Health vs Heart Disease



Insight

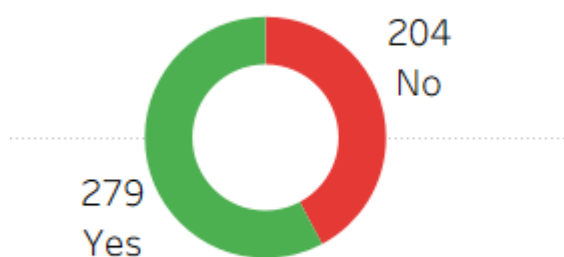
The chart analyzes the relationship between self-reported general health and heart disease. Individuals reporting poorer general health tend to show a higher occurrence of heart disease in the dataset. This highlights the importance of overall health in understanding heart disease trends.

Business Question 8

What is the impact of physical activity on heart disease?

Visualization

Physical Activity vs Heart Disease



Insight

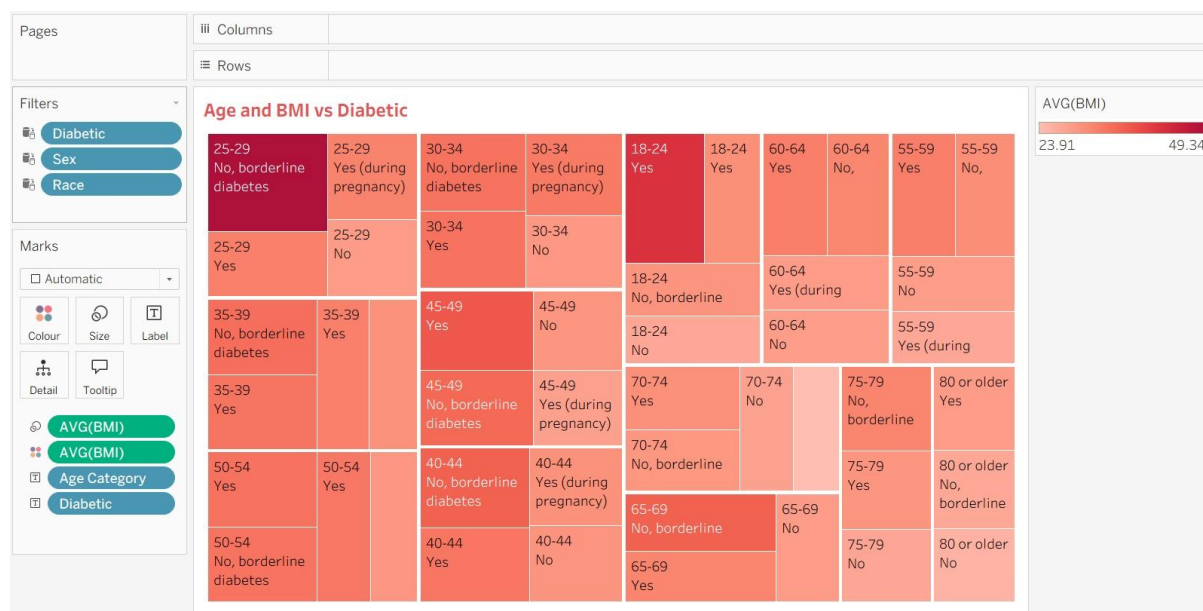
This visualization compares heart disease cases among physically active and inactive individuals. It helps understand the relationship between physical activity and heart disease occurrence. The comparison supports analysis of lifestyle factors that may influence heart health.

Business Question 9

How are age and BMI related to diabetes?

Visualization

Age and BMI vs Diabetic



Insight

The chart examines the relationship between age, Body Mass Index (BMI), and diabetes. It provides a visual comparison of diabetic cases across different age groups and BMI levels. This analysis helps identify patterns that may contribute to diabetes risk.

Business Question 10

How many people are affected by stroke, heart disease, and diabetes simultaneously?

Visualization

People got Stroke suffering from Heart Disease and Diabetic

Heart Disease Story



Insight

This visualization identifies individuals who are affected by multiple health conditions simultaneously. It helps recognize high-risk patient groups and provides a better understanding of overlapping diseases. Such insights are useful for comprehensive health analysis and informed decision-making.